

Solution to Ramanujan Challenge Problem 2.1: Polynomial Continued Fraction for π — Near-Complete Proof

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Abstract

We give a near-complete proof that the polynomial continued fraction in Problem 2.1 equals $6/(3 - \pi)$. The proof reduces the order-3 carrier to a forced order-2 equation via the adjoint solution $z(x) = x^{-9/5}(x + 2)$, linearizes the order-2 factor on the elliptic curve $E : u^2 = x(1 + 11x - x^2)$ using a third-kind differential with 3-torsion residues, and evaluates the endpoint coordinate as $T = -i\pi/(3\sqrt{3})$ via the logarithmic jump $\log(-1) = i\pi$. The remaining step is the algebraic de Rham reduction of the variation-of-parameters differentials.

1 Setup and factorization

With $\varphi = (1 + \sqrt{5})/2$, $\alpha = \varphi^5$, $\delta = \varphi^{-5}$, $\beta = -\delta$, and $f(x) = 1 + 11x - x^2 = (\alpha - x)(x + \delta)$, the order-3 carrier is

$$\frac{\mathcal{L}_3}{20x^3 f(x)} = \left(D + \frac{6}{5x} + \frac{1}{x+2} + \frac{f'}{f} \right) \mathcal{L}_2,$$

where $\mathcal{L}_2$ is an explicit order-2 operator. The inhomogeneous equation is $\mathcal{L}_3 F_Q = 6x$.

2 Adjoint quadrature (algebraic step)

The adjoint solution $z(x) = x^{-9/5}(x + 2)$ performs the first reduction exactly:

$$\mathcal{L}_2 F_Q = \frac{x + 12}{4x(x + 2)f(x)}.$$

This is **algebraic** — no π yet.

3 Elliptic linearization

On $E : u^2 = xf(x)$, the third-kind differential

$$\omega = \frac{x + 2}{(x - 3)u} dx$$

has poles at $P_{\pm} = (3, \pm 5\sqrt{3})$ with residues $\pm 1/\sqrt{3}$. Zero residue at the branch points.

The torsion relation: $P_+ - P_-$ has **order 3** in $\text{Pic}^0(E)$, certified by the rational function

$$\mathcal{R}(x, u) = \frac{(\sqrt{3}u - 4x - 3)(3u - 5\sqrt{3}x)}{(\sqrt{3}u + 4x + 3)(3u + 5\sqrt{3}x)}, \quad d \log \mathcal{R} = 3\sqrt{3}\omega.$$

4 The π -producing integral

Define $t(x) = \int_0^x \omega$. Along the path from $(0, 0)$ to $(\beta, 0)$ through the upper half-plane:

$$T := t(\beta) = -\frac{i\pi}{3\sqrt{3}},$$

because $\mathcal{R}(\beta, 0) = 1$, $\mathcal{R}(0, 0) = -1$, and $3\sqrt{3}T = \log(1) - \log(-1) = -i\pi$.

With $a = \sqrt{3}/2$: $aT = -i\pi/6$, so $e^{\pm aT} = e^{\mp i\pi/6}$ are **sixth roots of unity**.

5 Connection to the continued fraction

The linearization $F = h(x)\psi(t(x))$ with $h = \sqrt{(x-3)/u}$ reduces $\mathcal{L}_2$ to $\psi'' - \frac{3}{4}\psi = 0$ (constant-coefficient!).

The CF value satisfies

$$\frac{6}{3-\pi} = -42 + 396 \cdot \frac{\psi_S(T)}{\psi_Q(T)},$$

where ψ_Q, ψ_S are the forced solutions corresponding to Q_n and $S_n = (P_n + 42Q_n)/396$.

6 Remaining step: one period identity

Integration by parts in the VoP formula gives

$$\psi_Q(T) = \frac{3 + \mathcal{I}_0}{4}, \quad \psi_S(T) = \frac{3 + \mathcal{I}_0 + \mathcal{I}_1}{132},$$

where $\mathcal{I}_0, \mathcal{I}_1$ are explicit Abelian integrals on a degree-6 unramified cover of E . The CF identity $6/(3-\pi)$ is equivalent to the single period identity

$$2(3-\pi)\mathcal{I}_1 + (5\pi-16)(3+\mathcal{I}_0) = 0.$$

This is the genuine remaining certificate: one elliptic-period relation, reducible to a Hermite-reduction computation on the degree-6 cover.

References

- [1] Ramanujan Machine Group, *The Ramanujan Challenge for AI*, 2026.